UNIVERSITY OF LONDON

MSc EXAMINATION 2025

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS5860: Advanced Distributed Systems and Communication Networks

Time Allowed: **TWO hours**

Please answer **ALL** questions

Calculators are not permitted

©Royal Holloway, University of London 2025

Important Copyright Notice
This exam paper has been made available in electronic form
strictly for the educational benefit of current Royal Holloway students
on the course of study in question.
No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

1. (a) Which of the following statements are safety conditions, and which are liveness? **Briefly** justify your answer.

   - In a Chord DHT, no two nodes should be assigned the same identifier.
   - A newly joined node in Chord should not start serving lookups until it has correctly initialized its finger table.
   - The Chord protocol must guarantee that any key lookup eventually reaches the correct node responsible for that key.

   [3 marks]

   (b) Describe 2 differences and 1 commonality of the asynchronous model and the synchronous model in the context of message passing systems. [6 marks]

   (c) Let $v(e)$ denote the vector clock of event e, and $t(e)$ denote the Lamport logical clock of event e. If two vector clocks for events a, b are incomparable we write $v(a) <> v(b)$.

   Which of the following statements is/are true? **Briefly** justify your answer.

   - If $t(a) = t(b)$, then a and b must be concurrent.
   - If $v(a) < v(b)$ and $v(b) < v(c)$, then $t(a) < t(c)$.

   [6 marks]

   (d) Consider the space-time diagram in Figure 1.

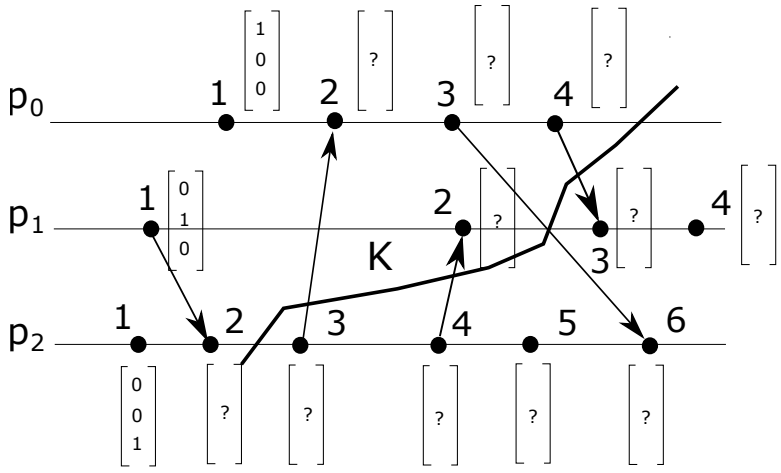


FIGURE 1: SPACE-TIME DIAGRAM WITH CUT K

   i. In the diagram, the first event at each process $p_0 \ldots p_2$ has been assigned a vector timestamp, but the vector timestamps for subsequent

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

events are blank (indicated with '?'). Create a diagram showing appropriate vector timestamps for all the events at each process. [6 marks]

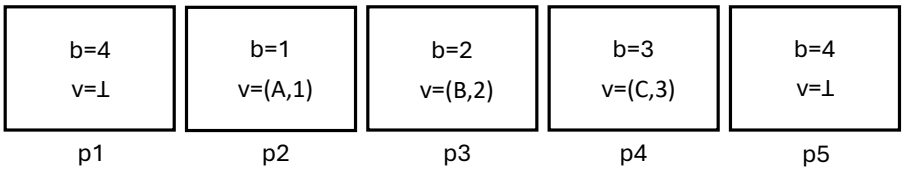
ii. In the *Recent Consistent Cut* problem, the processes are given a cut K and must determine the *maximal* consistent cut less than or equal to K ($\leq K$). Compute the maximal consistent cut that is less than or equal to ($\leq$) the cut K indicated in Figure 1. [6 marks]

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

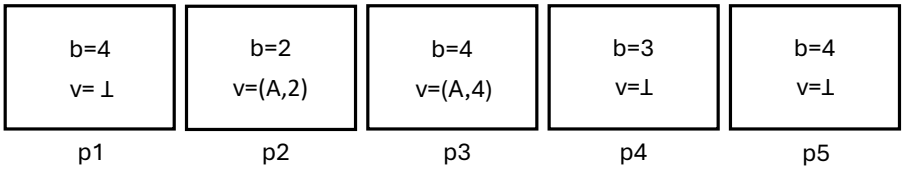
2. (a) Give the definition of 3 properties that any algorithm must satisfy if it solves consensus in the presence of crash failures. [6 marks]

(b) Consider a Paxos deployment of 5 nodes $p_1 \ldots p_5$, all working on the same consensus instance. Let b denote the last ballot number *promised* by a node, and $v = (v_{acc}, v^b_{acc})$ denote the last value *accepted* (v_{acc}) and the ballot in which it was accepted (v^b_{acc}). $v = \perp$ denotes no value was accepted yet.

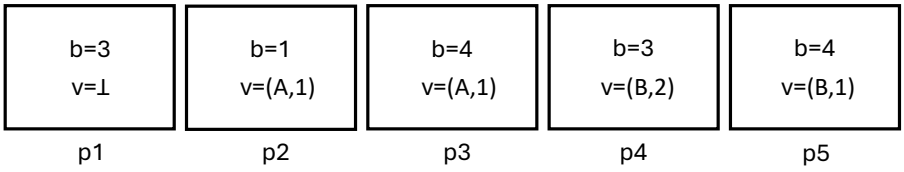
i. Given the following states for nodes $p_1 \ldots p_5$, what value or values can p_1, the leader propose in the next ballot (ballot 5)? Justify your answer. [2 marks]



ii. Given the following states for nodes $p_1 \ldots p_5$, what value or values can p_1, the leader propose in the next ballot (ballot 5)? Justify your answer. [2 marks]



iii. Is the following a possible state in a valid Paxos run? Why or why not? Has a value been decided yet if it is a valid state? [4 marks]

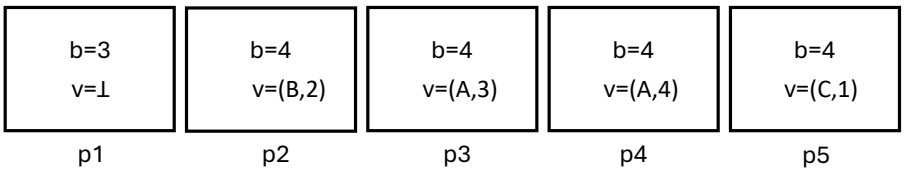


iv. Is the following a possible state in a valid Paxos run? Why or why not? Has a value been decided yet if it is a valid state? [4 marks]

(c) Consider the problem of distributed consensus in the Byzantine fault-tolerant (BFT) setting with n processes of which at most f are faulty.

i. In the synchronous model, what is the maximum number of faulty processes a correct BFT consensus algorithm can tolerate. [2 marks]

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.



ii. The Paxos algorithm for the crash-fault tolerant setting uses a quorum size of $2f + 1$. Would adapting the Paxos algorithm by simply increasing the quorum size be sufficient for it to represent a correct solution to the BFT consensus problem? Justify your reasoning. [4 marks]

(d) State the CAP theorem and describe its implications for building large web applications. [6 marks]

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

3. (a) Describe three ordering properties that can be required when delivering broadcast messages. [6 marks]

   (b) Pinster.net, a decentralized photo-sharing platform, allows clients to create new albums and upload photos. Each client participates in a peer-to-peer broadcast protocol to disseminate album creation and photo upload events to other users.

   To maintain a globally consistent user experience, Pinster.net must ensure that all users see album creation and photo upload events in the exact same order. This means that if one user sees album A created before album B, then all users must observe the same order. Similarly, if a user sees photo p_1 uploaded before p_2, every other user must see p_1 before p_2, regardless of whether the photos were uploaded by different users or to different albums.

   i. Define a suitable broadcast ordering guarantee to enforce these properties. Justify your reasoning. [4 marks]

   ii. Do you agree that these are suitable ordering properties for such a service to enforce? Justify your reasoning. [2 marks]

   (c) Consider the sequence of enqueue and dequeue operations shown in Figure 2. State whether the operations shown conform to the sequential specification for a first-in-first-out (FIFO) queue, where Q is an instance of a queue. You may assume the start of an interval indicates the invocation time of an enqueue/dequeue operation and the end of the interval represents the time of the response. **Briefly** justify your answer. [4 marks]

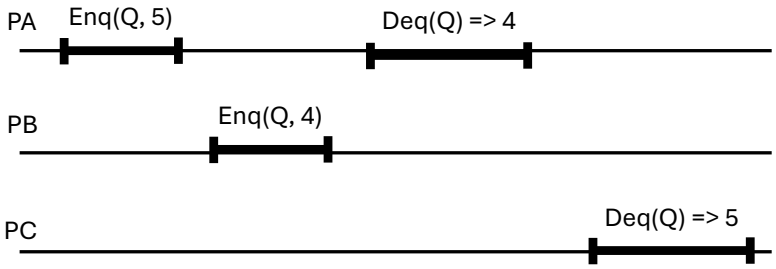


FIGURE 2: SEQUENCE OF ENQUEUE/DEQUEUE OPERATIONS

   (d) Consider the shared object execution sequence in Figure 3 for two processes p_A and p_B. Assume all writes are to the same shared object X and both processes read continuously. State whether the following operation orderings observed by p_A and p_B are linearizations of the execution sequence, sequentially consistent, both or neither. **Briefly** justify your answers.

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

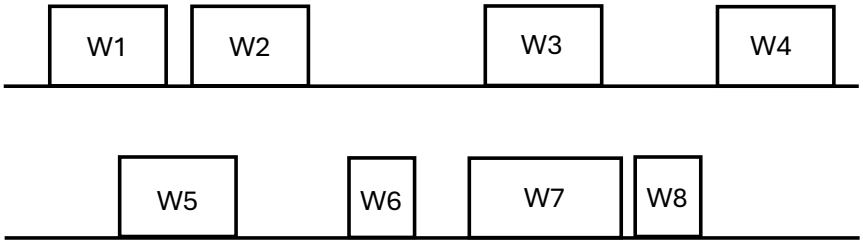


FIGURE 3: SHARED OBJECT EXECUTION SEQUENCE

i. p_A and p_B: W1,W2,W3,W5,W6,W4,W7,W8
ii. p_A and p_B: W1,W2,W4,W5,W6,W3,W7,W8
iii. p_A and p_B: W1,W5,W2,W6,W3,W7,W8,W4
iv. p_A: W1,W2,W3,W6,W4,W5,W7,W8, and
p_B: W2,W1,W4,W3,W6,W5,W7,W8

[8 marks]

NEXT PAGE

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

4. (a) What is a double-spend attack in the context of crypto-currencies? Under what circumstances could a double-spend attack occur on the Bitcoin blockchain and how can Bitcoin users protect themselves against such attacks? [6 marks]

   (b) In the context of storage in peer-to-peer networks, suppose that you are given an ID-cycle of size $32 = 2^5$, where the vertices are numbered in the order $0, \ldots, 31$. The following nodes are connected according to the edges of the ID cycle: p_0 with ID 5, p_1 with ID 12, p_2 with ID 18, and p_3 with ID 29.

      i. Create a diagram of the ID-cycle above showing clearly the position of each node on the ring. [2 marks]

      ii. Assuming appropriate shortcut connections have been set up in order to establish a Chord topology, list the hops taken for a $put_i(k)$ operation for key $k = 4$ starting from process $p_i = p_1$. [7 marks]

      iii. Chord finger table routing routing was originally designed for large-scale peer-to-peer filesharing systems. Discuss its disadvantages in a modern data centre setting (e.g. as part of a distributed key-value store) and why its use of consistent hashing is still relevant. [4 marks]

END

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.